University of Chittagong

7th Semester B.Sc (Engg.)Examination 2019

Department of Computer Science & Engineering

Course Code: CSE 715

Course Title: Machine Learning

Time: 4 Hours **Full Marks: $52\frac{1}{2}$**

[Answer six questions taking any three from each section and use separate answer script for each section]

Section A

1. a) The Apriori algorithm makes use of prior knowledge of subset support properties. 3
   i) Prove that all nonempty subsets of a frequent itemset must also be frequent.
   ii) Prove that the support of any nonempty subset s' of itemset must be at least as great as the support of s.

   b) Apriori and FP-Growth are two famous algorithms for frequent itemset generation. Compare the efficiency pf the two mining process. 1.75

   c) A database has eight transactions. Let min_sup=2. Find all frequent item sets using Aprori Algorithm. 4

TID	1	2	3	4	5	6	7	8
Items	K,C,A	A,M,D	C,K,D	K,D,A	A,C,S	D,A,S	K,S,A	A,M,K

2. a) State why Random Forest may improve the accuracy of decision tree induction. 2.75

   b) Consider the table I for a binary class problem. 6

Table I

A	B	Class Label
T	F	+
T	T	+
T	T	+
T	F	-
T	T	+
F	F	-
F	F	-
F	F	-
T	T	-
T	F	-

   i. Calculate the information gain when splitting on A and B. Which attribute would the decision tree induction algorithm choose?
   ii. Calculate the gain in the Gini index when splitting on A and B. Which attribute would the decision tree induction algorithm choose?

3. a) Bayesian classifiers are statistical classifiers. Justify your answer. 1.75
   b) Why naive Bayesian classification is known as class-conditional independence? 2
   c) Suppose playing record of tennis is shown in table II. 5

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Table II

	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Today is Sunny, Mild, High and Strong. Using Naïve Bayesian Classifier determine if anyone will play tennis today.

4. a) It is difficult to assess classification accuracy when individual data objects may belongs to more than one class at a time. In such cases, comment on what criteria you would use to compare different classifiers modeled after the same data. 2.75

b) Suppose that we would like to select between two prediction models M1 and M2. You need to smooth both model's Recall-Precision (R-P) using interpolation between 0~1, and also need to draw R-P curve for model selection. 3

M1					
Recall	0.20	0.40	0.70	0.90	1.0
Precision	0.12	0.40	0.67	0.75	0.88

M2					
Recall	0.10	0.30	0.86	0.96	1.0
Precision	0.19	0.45	0.76	0.85	0.98

c) Construction and evaluation of a classifier require partitioning labeled data into a training set and a test set. Sketch the following methods for the following data set. 3

i) Cross Validation (k=2)

ii) Stratified Cross Validation (k=2)

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tid	Dimension1	Dimension2	Dimension3	Dimension4	Class Label
1	A1	B1	C1	D1	E1
2	A1	B1	C1	D2	E1
3	A1	B2	C1	D1	E1
4	A1	B3	C2	D1	E2
5	A2	B1	C1	D1	E1
6	A2	B3	C2	D2	E1
7	A4	B2	C1	D1	E2
8	A4	B3	C2	D1	E2
9	A4	B3	C2	D2	E1
10	A4	B2	C2	D1	E2

Section B

5. a) Suppose that the machine learning task is to cluster the following students CGPA into three clusters: 4.75

A	B	C	D	E	F
5.45	3.88	2.78	1.25	2.2	3.80

The distance function is Manhattan distance. Suppose initially we assign B, D, and F as the center/representative of each cluster, respectively. Use k-means algorithm to show only the final three clusters.

b) Use the following dissimilarity matrix to perform AGNES. Show you results by drawing a dendogram. The dendogram should clearly show the order in which the points are merged. 4

	P1	P2	P3	P4	P5
P1	0.00	0.10	0.41	0.55	0.35
P2	0.10	0.00	0.64	0.47	0.98
P3	0.41	0.64	0.00	0.44	0.85
P4	0.55	0.47	0.44	0.00	0.76
P5	0.35	0.98	0.85	0.76	0.00

6. a) How can you improve the interpretability of the learned network in backpropagation? 2
   b) Why neural network learning is referred as connectionist learning? 2
   c) Why we use a squashing function in backpropagation? 1.75
   d) For the following table, design a multilayer feed-forward neural network and show weight values after one iteration of the backpropagation algorithm, given the training instance X=(**Body Temperature=warm, Gives Birth=yes, Four-legged=no, Hibernates=yes).** Indicate your initial weight values and biases and the learning rate used. 3

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Body Temperature	Gives Birth	Four-legged	Hibernates	Class Label
Cold	no	Yes	yes	no
Cold	yes	No	no	no
Warm	no	No	no	no
Warm	no	No	yes	no
Warm	no	Yes	yes	yes

7. a) Illustrate the limitation of k-means algorithm. 3
   b) Use an example to show why k-means algorithm may not find the global optimum that is optimizing the within-cluster variation. 2.75
   c) Clustering is recognized as an important datamining task with broad applications. Give one application example that uses a clustering as a preprocessing tool for data preparation for other data mining tasks. 3

8. a) Support Vector Machine always tries to maintain maximum margin for classification. Explain it with proper sketch. 2.75
   b) What would be the best generalization for choosing the number of support vectors, if the dimensionality of data is high? 2
   c) Suppose we are given the following positively labeled data points in 2D space: {(4,1), (4,-1), (5,1), (5,-1)} and the following negatively labeled data points in 2D space: {(2,0), (0,2), (0,-2), (-2,0)}. In addition, three support vectors are {(2,0), (4,1), (4,-1),} Apply Support Vector Machine to classify data objects (6,-1). 4

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